Dynamic Trader - Architecture Guide

This document explains every major part of the cod

Related: DT.md (CLI cheat sheet), README .md (rese

1. Big picture

```
OKX daily trade zips
```

|

Rule of thumb

Concern | Language | Why

Tick tape, fil

data/ is gitignored. Archives and scan o utputs sta

2. Repository layout

```
ppo-trader/
```

```
+-- Cargo.
```

3. Rust library modules (src/)

Module | Role

okx_history.

Fast path vs Nautilus

1. Fast path: Tape::load -> scan::simula

te / tpsl_g

Feature verify (default on) pulls in nau

tilus-*. En

```
cargo build --profile relfast --no-default-features --bi
n download
```

4. Rust binaries (src/bin/)

Binary | Command | Needs Nautilus? | What it does

download | d

dt price-walk execs window_nm_walk_forwa

rd --path p

Scan on-disk format

Window bucket (--bucket window, default)

- meta.json - timeframes, lengths, TP/SL

/leverage,

Month bucket (--bucket month)

- month_YYYY-MM.bin / all.bin - 12-byte

records (s

Go dt browse auto-detects layout from me

ta.json.

simulate JSONL contract

Request line:

```
{"id":"1","minutes":60,"fast":14,"slow":50,"exits":{"sto
p_pct":1.0
```

Response line: outcome, optional closed

trades, op

5. Go product (cmd/dt, internal/)

Build on macOS:

```
./scripts/build_dt.sh
```

Command | Behaviour

dt download

Score model (internal/score)

Port of src/window_score.rs:

- Features in [0,1]: winrate, gain, R/R,

until, dra

Walk-forward score path (internal/walkfo

rward)

1. Read meta.json + window bins -> final

ists

Unit tests live under internal/score and

internal/w

6. Typical research workflow

```
# 1. Archives (2 years, end exclusive)
```

cargo run

```
# 2. Window scan (staged TP/SL grid -> survivors)
```

cargo run

```
# 3. Browse results
```

./scripts/

```
# 4. Score walk-forward (Go)
```

./dt walk-

```
# 5. Price-path walk-forward (still Rust)
```

```
./dt price
```

7. Config & safety

config/testnet.toml:

- risk.paper_only must be true

- risk.max

Secrets only in .env (gitignored). Never

commit k

8. What was removed in the cleanup

Deleted as unused after the split (Go ow

ns produc

- browse_full_scan / browse_window_scan

(+ HTML)

Still present for engine / price path: d

ownload,

9. Mental model for reading code

1. Data in: okx_history + tape

2. Signal:

When changing fill rules, change Rust sc

an + strat